Thus

Numerically, is very small. As above, is approximately given by

Subtract to find satellite minus ground clock readings:

Solve for and substitute into :

The stated is ground elapsed time . Finally, after , the satellite minus ground reading is

3.

. For constant room temperature, the heat pump output must equal the room heat loss rate. Let be the indoor heat output and the electric input. The heat extracted outdoors at low temperature per unit time is . Then

is the indoor hot absolute temperature, the outdoor one. From ,

Minimum cost requires minimum : equality in " " in , the ideal case, minimizes . Thus the least electric power is

Heat conducts out of the room through the glass. The loss rate is

For a constant room temperature,

Equations give